

SMoDO - Practical 11 - Stata

Dataset required: `SAheart.csv`

In this practical we will explore ridge and lasso regression using the South Africa data used in the previous practical.

1. Load the dataset in Stata and fit the same logistic regression model as in the previous practical for the `chd` outcome, with covariates `sbp tobacco ldl famhist obesity alcohol age`, saving the estimates into an object called `logit`:

```
logit chd sbp tobacco ldl i.famhist obesity alcohol age
estimates store logit
```

2. We will next fit the model using ridge regression. While Stata has a separate command for lasso, it does not for ridge regression. It does however have `elasticnet` for elastic net regression, which can be used to run ridge regression by specifying the α parameter which controls the linear combination of quadratic and absolute value penalty.

- (a) Examine the help for `elasticnet` and find the value of alpha needed for ridge regression. Then complete & use the following code to run the ridge regression with 10-fold CV and save the estimates:

```
elasticnet logit chd sbp tobacco ldl i.famhist obesity alcohol age /*
*/, alpha( ) rseed(723963)
estimates store ridgecv
```

According to the CV, what value of λ was found to be optimal?

- (b) Examine how model performance varies with λ by running:

```
cvplot
```

What do you conclude based on the plot?

- (c) Examine how the coefficients vary with the choice of λ , using

```
coefpath, xunits(lnlambda) legend(on)
```

What do you notice about the family history variable?

- (d) Run the following code to compare the coefficients from the standard logistic compared to the ridge regression:

```
lassocoef logit ridgecv, display(coef)
```

What do you notice about the family history variable? To understand what's going, load the help for `lasso`, click to load the PDF manual, and read pages 36 and 37.

- (e) The default for displaying coefficients using `lassocoeff` from penalized regression methods is to show coefficients for the standardized variables. To compare like with like, run:

```
lassocoeff logit, display(coef)
lassocoeff ridgecv, display(coef, penalized)
```

Compare the estimated coefficients between logistic and ridge logistic regression – are they in line with what you expect?

- (f) To further explore the impacts of shrinkage, generate variables with model fitted probabilities from the ordinary logistic model and the ridge logistic model. Then summarise the fitted probabilities – what do you notice?

3. We will now explore lasso regression.

- (a) Run lasso regression, using the same seed as we did earlier for ridge:

```
lasso logit chd sbp tobacco ldl i.famhist obesity alcohol age /*
*/ , rseed(723963)
estimates store lassocv
```

How many covariates has lasso regression selected/dropped?

- (b) Run the following code to look at the coefficients and how they change with λ :

```
coefpath, xunits(lnlambda) legend(on) xline(.0190284)
```

Based on the plot, which covariates have been retained? How well do the covariates dropped correspond with the p-values from the standard logistic regression ran right at the start?

- (c) Run the following code to look at the coefficient estimates (on the original variable scales):

```
lassocoeff ridgecv lassocv, display(coef, penalized)
```

What has happened with the family history variable?

- (d) Thus far we have been choosing λ based on the deviance (log likelihood) and cross-validation. But there are other alternatives. One that is available in Stata is to choose λ using the Bayesian Information Criterion (BIC). This is the log likelihood with a penalty for how many parameters the model contains. This does not require use of CV – larger models have higher log likelihood, but they are penalized for this because they have more parameters.

Run lasso with BIC used to select λ :

```
lasso logit chd sbp tobacco ldl i.famhist obesity alcohol age/*
*/ , selection(bic)
estimates store lassobic
lassocoeff ridgecv lassocv lassobic, display(coef, penalized)
```

How does the value of λ selected compare with when we used CV and the deviance/log-likelihood? Does the change in selected λ have the expected impact?